

FIRST SESSION – TENTH PARLIAMENT

REPORT

OF THE PORTFOLIO COMMITTEE ON ENERGY AND POWER DEVELOPMENT

ON

THE STATE OF ELECTRICITY SUPPLY IN ZIMBABWE

SC02

ORDERED IN TERMS OF STANDING ORDER No. 18:

1. At the commencement of every session, there shall be as many Committees to be designated according to government portfolios as the Committee on Standing Rules and Orders Committee may deem fit.
2. Each Select Committee must be known by the portfolio determined for it by the Committee on Standing Rules and Orders.

TERMS OF REFERENCE OF PORTFOLIO COMMITTEES S.O 21

Subject to these Standing Orders a Portfolio Committee must:

- Examine expenditure administration and policy of government departments and other matters falling under their jurisdictions as Parliament may, by resolution determine;
- Consider and deal with all Bills and Statutory Instruments or other matters which are referred to it by or under a resolution of the House or by the Speaker;
- Consider or deal with an appropriation or money bill or any aspect of an appropriation or money bill referred to it by these Standing Orders or by or under resolution of this House;
- Monitor, investigate, enquire into and make recommendations relating to any aspect of the legislative programme, budget, policy or any other matter it may consider relevant to the government department falling within the category of affairs assigned to it, and may for that purpose consult and liaise with such a department; and
- Consider or deal with all international treaties, conventions and agreements relevant to it, which are from time to time negotiated, entered into or agreed upon within four weeks.

SPEAKER'S ANNOUNCEMENT

On Wednesday, 18 October 2023, The Speaker announced that the Committee of Standing Rules and Orders had appointed Members to serve in the Portfolio Committee on Energy and Power Development is as follows:

Hon. Butau Dzidzai, Hon. Chari R., Hon. Chigumbu D., Hon. Chokururama J., Hon. Dube P., Hon Guyo P., Hon. Makumbe T., Hon. Mambipiri G., Hon. Maphosa W., Hon Marashe S., Hon. Mapiki J., Hon. Marange N., Hon. Mhangwa L.E., Hon. Musiiwa R., Hon. Nyathi T., Hon. Moyo F., Hon. Jonga W. A., Hon. Mukungunugwa G. H., Hon. Ndlovu B., Hon. Samambwa., Hon Sibanda Libion., Hon. Shumba T., Hon. Taruvinga F., Hon. Tshuma S.

Hon. Sihlabo. V—to be Chairperson.

1.0 INTRODUCTION

Flavi (1986) asserts that the real potential of electricity lies not only in improving social amenities but in stimulating long term economic growth. . Electricity is an enabler in the development matrix and economic growth for any country. Zimbabwe's power generation primarily relies on hydropower and coal. Zimbabwe has been facing power outages for the past years due to a combination of factors which include aging infrastructure, inadequate investment and climate change amongst other factors. In response to this crisis, the Parliamentary Portfolio Committee on Energy and Power Development initiated an enquiry to assess the state of electricity supply. This report presents the findings, observations, and recommendations of that inquiry, delving into the challenges, root causes, and consequences of Zimbabwe's electricity shortage as well as proffering recommendations.

2.0 OBJECTIVES

The objectives of the enquiry were to:

1. Establish the current capacity of power generation within the country;
2. Assess the overall state of electricity supply and distribution nationwide; and
3. Identify and analyse the challenges hindering the efficient generation, transmission and distribution of electricity and propose appropriate recommendations.

3.0 METHODOLOGY

The Committee held oral evidence sessions with Rural Electrification Fund, Zimbabwe Electricity Supply Authority Holdings and its subsidiaries and the Zimbabwe Energy Regulatory Authority in order to get a first-hand understanding of the power generation, electricity supply and distribution in the country and challenges that are being faced. The Committee then conducted site visits to power generation plants and transmission infrastructure namely Kariba power station, Sherwood, Gwanda and Hwange power station.

4.0 COMMITTEE'S FINDINGS

4.1 LEGAL FRAMEWORK GOVERNING THE ELECTRICITY SECTOR

ZESA management highlighted that electricity sector is mainly governed by the Electricity Act [Chapter 13:19] and the Energy Regulatory Authority Act [Chapter 13:23]. These two acts provide the legal framework for the generation, transmission, distribution, and supply of electricity in Zimbabwe. Therefore, the Committee noted that the Electricity Act establishes

the Zimbabwe Electricity Regulatory Commission, which is responsible for regulating the sector and ensuring compliance with standards and regulations however, there are gaps that are found in this Act. The Committee cited that the act does not address the need for modernizing and expanding electricity infrastructure which is the major challenge in the electricity sector.

In addition, the act lacks detailed provisions to promote the adoption of renewable sources such as solar and wind and it also does not address the issue of tariff setting that affects the financial viability of power generation projects. Committee also noted that the Act enforcement measures for harsher penalties that were imposed for vandalism and theft

ZESA management mentioned that the Energy Regulatory Authority Act, on the other hand, empowers the Zimbabwe Energy Regulatory Authority (ZERA) to issue licenses, regulate tariffs, and enforce compliance with regulatory requirements. The Committee then noted that, this act could be enhanced its effectiveness if some of the areas are improved that include clarity on renewable energy promotion. The Committee added that the act mentions renewable energy but there are no specific measures and incentives for the renewable energy development. Furthermore, members pointed out that the act could improve on the resolution of disputes between regulated utilities and their customers.

4.2 Current Capacity of Power Generation

The Committee visited the Kariba South Power Station (KSPS) and was informed by the Zimbabwe Power Company (ZPC) management that Kariba Hydroelectric Complex has a storage capacity of 64.74 billion cubic meters for electricity generation and there are two power stations on the complex, that is, Kariba South on the Zimbabwean side and Kariba North on the Zambian side.

ZPC added that KSPS is capacitated to produce 1050Megawatts (MW) but currently is only producing 250MW. Management stressed out that the normal year water allocation is 20 billion cubic meters and allocation for the year 2024 was 8 billion cubic meters. Lake level was at 477.4 meters as of May 2024, resulting in the production of only 250 MW. The Committee noted that this significant decrease in water allocation for 2024 has raised concerns about the sustainability of current energy production levels.

In addition, the Committee also visited Hwange Thermal Power Station. ZPC emphasized that the power station serves as the nation's base load station and the features of the power station ensure a steady supply of power at all times. ZPC explained that the power station is capacitated

to produce 1520MW inclusive of the 2 units namely Hwange 7 and 8 that were commissioned in 2018 providing 300MW each. It was further explained that these 2 units are a contributor to the self-sufficiency in electricity generation helping in the reduction of importation of electricity and help to retain foreign currency.

In addition, ZPC mentioned that Independent Power Producers (IPPs) are playing a crucial role in the energy sector as most of them use various energy source such as solar helping in diversifying the energy mix. Management added that the IPPs are also contributing 69 MW to the national grid.

PLANTS	CAPACITY IN MEGAWATTS (MW) (When at 100%)	ELECTRICITY GENERATED IN MEGAWATTS (MW) (currently)	SHORTFALL
Kariba South Power Station	1 050	250	800 MW
Hwange Power Station	1 520	760	760 MW
Independent Power Producers		69	
TOTAL	**2 570**	**1079**	**1 560**

Fig 1: Combined Installed Capacity of Power Plants

4.3 Overall State of Electricity Supply and Distribution

The Committee also visited the Sherwood transmission hub and assessed the old Sherwood transmission substation in Sherwood as well as the Hwange 7 and 8 transmission substation, also known as Sherwood B. Zimbabwe Electricity Transmission and Distribution Company (ZETDC) responsible for the operation and transmission, network rehabilitation, network expansion and project management informed the Committee about the new transmission lines being built, including the Alaska-Karoi. Committee commended the good works being done by ZETDC.

However, in as much as the transmission and distribution lines are functioning well, ZESA management highlighted that Zimbabwe is still facing serious power outages because the distribution supply of electricity is not meeting the nations demand. The Committee noted that the major challenge is of all power plants not producing their optimum capacity.

In addition, ZPC pointed out that KSPS is producing 250MW only, Hwange power station is producing 750MW and IPPs are adding 69MW making a total of 1079MW which cannot sustain the whole nation yet the nation's demand is from 1500MW to 2350MW. Furthermore, ZPC has resorted to importing power from the neighboring countries like Namibia in order to reduce load shedding. The Committee cited that considering the state we are in as a country, if there are no strategies that are going to be implemented sooner to boost generation, the nation will be suffering for a long haul.

4.4 Challenges hindering the efficient Generation, Transmission and Distribution

1. **Climate changes or variability**

   ZESA management cited that as a result of the climatic shift that has negatively impacted the water levels at the largest man-made lake, dam levels have been dramatically reduced, affecting the quantity of electricity produced by hydropower facilities. As a result, water allocations to KSPS are only 8 billion cubic meters, resulting in 250MW of output. This suggests that the country's power generation would be significantly reduced as water levels fall.

2. **Aging Infrastructure**

   ZPC pointed out the outdated infrastructure of the Kariba and Hwange power plants, which were built approximately 6 and 3 decades ago, respectively, posing operational issues. For example, the breakdown of KSPS Unit 4 in July 2022 emphasizes the importance of extensive maintenance modifications.

3. **Theft and vandalism**

   ZETDC reported substantial theft and vandalism issues, which have a negative impact on their business. This involves the theft of power cables, transformers, and other equipment, as well as the vandalism of infrastructure like power lines and substations. They said that these illicit operations cause service delays, financial losses, and increased maintenance expenses for ZETDC. Furthermore, ZETDC stated that they are working on how to replace copper wires that are attracting criminals, and that there are

rumours that the theft and vandalism are an inside job, but there is no proof to support this.

4. **High foreign currency obligations for loan repayments**

   According to ZESA Holdings, they possess huge loan burden in renovating some of the the units KSPS which amounts to $350 million. ZESA created a huge debt burden on the utility, including foreign currency commitments for loan repayments and spare parts purchase. Management added that repayments need to be done in foreign currency and they have resorted to exporting power in order to generate foreign currency.

5. **Skills flight**

   ZESA management added that there is shortage of skilled personnel in the electricity sector resulting to lack of technical expertise and they cited that over last 2 years, they have struggled to retain critical skills for the efficient running of the electricity generation plants. ZETDC mentioned that they are enrolling their own for technical expertise but mainly males are participating yet they are trying to break the gender stereotype.

7.0 COMMITTEE'S OBSERVATIONS

7.1 The Committee observed that Zimbabwe's electricity supply strategy has been overly reliant on the Kariba South Power Station upgrade, a short-term solution vulnerable to water constraints. This approach has failed to address the country's persistent electricity woes.

7.2 Furthermore, global funding trends favouring renewable energy projects have limited Zimbabwe's options for a balanced energy mix. The Committee observed the need for baseload power sources like thermal and nuclear plants to ensure stable electricity supply, while acknowledging the role of renewables in peak power generation.

7.3 Committee highlighted the country's history of poor loan servicing has hindered access to affordable financing for power projects. The Committee noted that the Hwange and Kariba projects were procured at inflated costs contributing to high electricity tariffs.

7.4 The Committee raised concerns about ZERA's independence in determining tariff structures in line with ZESA Holdings objectives. Additionally, the shortage of foreign

currency to service power project obligations and the ongoing brain drain in the energy sector pose significant challenges.

7.5 The Committee observed the importance of timely maintenance of transmission and distribution infrastructure to prevent system failures.

8.0 COMMITTEE'S RECOMMENDATIONS

Based on the above observations, the Committee made the following recommendations that focus on three broad areas namely; addressing infrastructure and energy mix, financial sustainability and governance and human capital and revenue generation:

8.1 ZESA Holdings should be actively involved in attracting significant investments in refurbishing existing power plants and constructing new ones to overcome the overreliance on Kariba South and address the observed infrastructure deficiencies by 31st December, 2025.

8.2 The Ministry of Energy and Power Development should diversify the energy mix to mitigate the impact of water constraints on hydroelectric power and enhance overall system reliability by 31st December 2026. In this regard, the Committee recommends for a strategic balance between base load power sources such as thermal and nuclear and renewable energy like solar, wind and geothermal.

8.3 Treasury should timely disburse funds to service loans for power projects to ease Zimbabwe's history of debt servicing challenges and improve the country's creditworthiness and facilitate access to affordable energy financing.

8.4 The Ministry of Energy and Power Development should invest in renewable energy projects in order to complement private sector efforts and address the funding constraints observed by 31st December 2025.

8.5 The Ministry of Energy and Power Development should implement stringent contract management protocols thorough due diligence, clear performance indicators and effective

dispute resolution mechanisms to prevent recurrences of the Gwanda Solar Project debacle by 31st December 2024.

8.6 The Ministry of Energy and Power Development and the Ministry of Finance, Economic and Development Investment Promotion should make deliberate efforts to retain existing talent and attract skilled Zimbabweans in the diaspora to address the critical shortage of skilled personnel in the energy sector by 31st December 2026.

8.7 The Ministry of Energy and Power Development should implement robust revenue collection mechanisms, by 31st December 2025, to ensure the financial sustainability of the power sector and support ongoing projects as well as contributing to reducing the government's financial burden on the sector.

8.8 ZPC and energy contractors should develop an effective communication and collaboration strategy for successful project implementation as a forward-thinking approach that prioritises the national interest that maximises project benefits by 31st December 2025.

9.0 CONCLUSION

Zimbabwe's power generation crisis poses a significant obstacle to economic growth. The escalating impacts of climate change underscore the urgent need for substantial investments in infrastructure and renewable energy sources. Fortunately, Zimbabwe is endowed with abundant renewable resources, including solar and hydropower, which offer a promising path toward addressing the country's energy supply challenges. By harnessing these resources, Zimbabwe can enhance its energy security, reduce reliance on fossil fuels and stimulate economic development.